

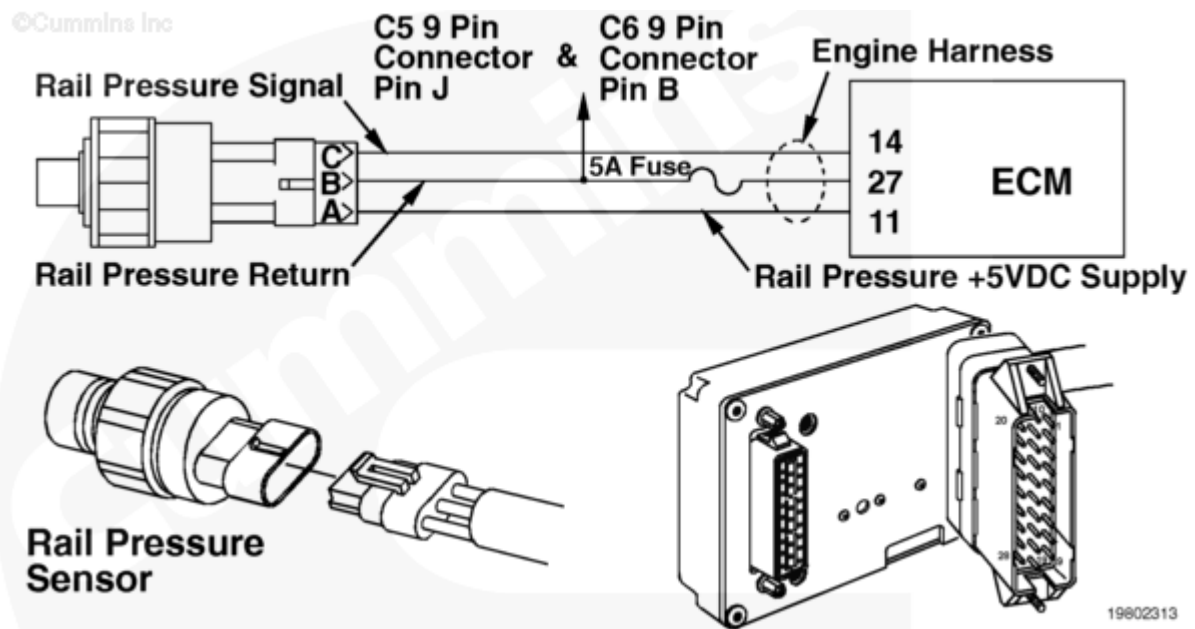
Fault Code: 451

Injector Metering Rail 1 Pressure Sensor Circuit - Voltage Above Normal or Shorted to High Source

 Printable Version

Overview

Codes	Reason	Effect
Fault Code: 451 PID(P): P94 SPN: FMI: 3 Lamp: On SRT: 00-640	High signal voltage detected at the rail pressure sensor circuit.	Power and engine speed derate.



Rail Pressure Sensor Circuit

Circuit Description

The rail pressure sensor is a variable resistance sensor used by the ECM to monitor the fuel rail pressure. This sensor has three circuits: 5-volt supply, return and signal circuits. This rail pressure sensor signal voltage changes based on the pressure in the fuel rail.

Component Location

The rail pressure sensor location may vary and is dependent on the engine.

Conditions For Running The Diagnostics

This diagnostic runs continuously when the keyswitch is in the ON position.

Conditions For Setting The Fault Codes

The Engine Control Module (ECM) detected the rail pressure signal voltage was out of range high.

Action Taken When The Fault Code Is Active

- The ECM illuminates the amber CHECK ENGINE light when the diagnostic runs and fails.
- Engine torque and/or engine speed derate will be active.

For Power Generation Applications:

- The generator set controller displays the fault immediately when the diagnostics runs and fails.
- Engine torque and/or engine speed derate will be active.

Conditions For Clearing The Fault Code

- To validate the repair, perform a key cycle, start the engine and let it idle for 1 minute.
- The fault code status displayed by the recommended Cummins® electronic service tool or equivalent will change to INACTIVE immediately after the diagnostic runs and passes.
- The ECM will turn off the amber CHECK ENGINE lamp immediately after the diagnostic runs and passes.
- The "Reset All Faults" command in the recommended Cummins® electronic service tool or equivalent can be used to clear active and inactive faults.

For Power Generation Applications:

- To validate the repair, start the engine and let it run for 1 minute at no load.
- The generator set controller will turn off the indicator immediately after the user presses reset.
- The "Reset All Faults" command in the recommended Cummins® electronic service tool or equivalent can be used to clear active faults.

Shoptalk

Possible causes of this fault code include:

- Malfunctioning or damaged engine wiring harness.
- Malfunctioning or damaged OEM wiring harness.
- Damaged or loose connectors.
- Malfunctioning or damaged rail pressure sensor.

Refer to Troubleshooting Fault Code t05-451 (</qs3/pubsys2/xml/en/procedures/98/98-t05-451.html>)

Last Modified: 15-Sep-2021
